

Abstract

Low Earth Orbit (LEO) satellite constellations are emerging as a key enabler for global Internet of Things (IoT) com

1. Introduction

Satellite IoT is important because it provides coverage to remote areas where terrestrial networks are unavailable.

2. Related Work

Prior work on satellite resource allocation includes fixed TDMA/FDMA schemes and convex optimization based po

3. System Model

We consider a Walker constellation with 60 LEO satellites at 550 km altitude and 53 degree inclination. Each sate

4. Problem Formulation

The joint resource allocation and access control problem is formulated as a mixed-integer nonlinear optimization.

5. Proposed Method

We propose a deep reinforcement learning algorithm based on the PPO framework. The state space includes cha

6. Algorithm

The proposed algorithm iteratively samples trajectories, estimates advantages, and updates the policy network. Th

7. Experiments

We simulate 60 LEO satellites and 300 IoT terminals over 10000 time slots. The carrier frequency is 20 GHz and t

8. Conclusion

This paper proposed a DRL-based joint resource allocation and access control scheme for LEO satellite IoT. The

References

- [1] Author A. et al., "Satellite IoT resource allocation: a survey," IEEE Journal, 2024.
- [2] Author B. et al., "Deep reinforcement learning for LEO networks," IEEE Transactions, 2023.